COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	9/5/2026	**DATE SUBMITTED:**	9/5/2026
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ACTIVITY NO. 7.1 Using Sorting Algorithms

1. PROCEDURE OUTPUT

Objectives: To create a program in C++ using sorting algorithms.

- 1. To be familiarized with the sorting algorithms.
- 2. To be able to different the types of sorting algorithms.
- 3. To be able to create a program with sorting algorithms.

Answer the following questions:

1. What is a sorting algorithm?
The sorting algorithm can be described as a procedure used in the arrangement of data in a specific order, including the order of ascending or descending . The main goal of the algorithm is to ensure that data is well organized for understanding it easily.

Example: The numbers 5, 2, 8, 1, 3 can be arranged in ascending order as 1, 2, 3, 5, 8.

2 .Where can sorting algorithms be used?
Sorting algorithms can be applied in various programs where organization of data is required. They are mostly utilized in educational institutions, organizations, shopping sites, banking software, and various other computer-based programs where there is the need to sort out large volumes of data.

Example: A school management program may apply sorting algorithms to sort out the students according to their grades from the highest to the lowest.

3. Explain the different types of sorting algorithms.
There are several algorithms that can be used to sort elements and these algorithms use different approaches to sort elements. Selection Sort uses the approach of finding the smallest element in the list and sorts it, while Insertion Sort uses the approach of selecting an element and sorting it with respect to other sorted elements already in the list. Bubble Sort uses comparison of elements to make any required swap between them.

Selection sort algorithm

- It finds the smallest element for the unsorted data of a list and places it at the current position of the compared data value. iIt begins with the first position, finds the minimum value from the remaining array, and swaps that with the first value. It continues to the next position and does the same until the whole array is sorted.

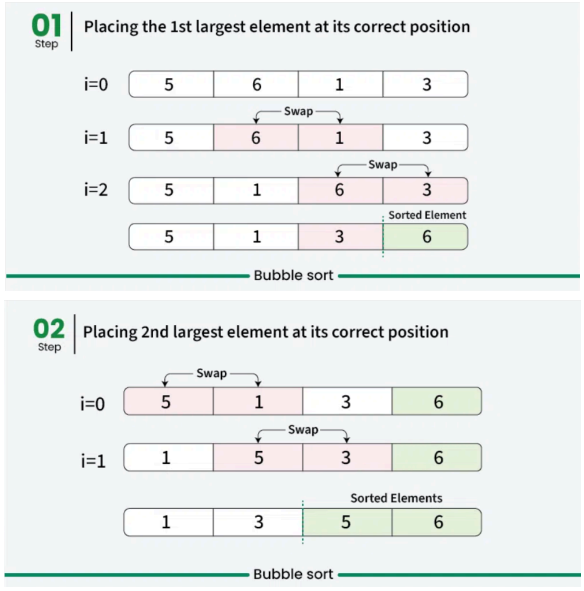
Insertion sort algorithm

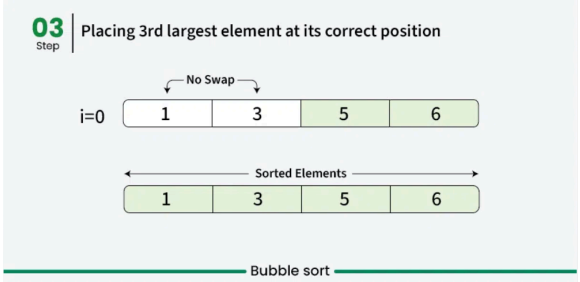
- It works by picking an element one at a time and putting them into their proper place amongst the already sorted elements. The way this is done is by beginning from the second element and comparing it to the elements ahead of it. If the elements ahead are larger than the selected element, they are shifted one place to the right and then the selected element is put in its place.

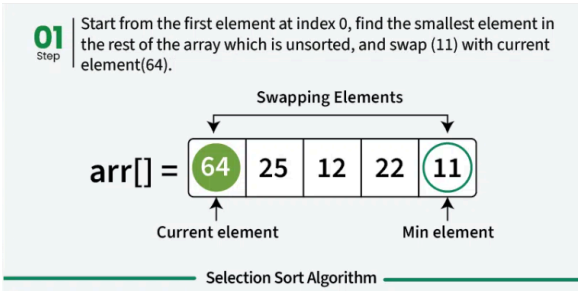
Bubble sort algorithm

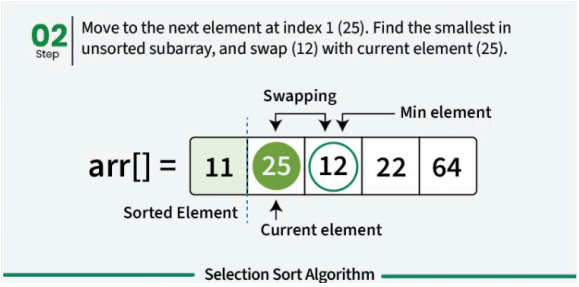
- Bubble Sort algorithm compares the 2 values iterating between the 2 for loop functions. If the outer loop commonly known as the variable **[i]** is greater than the inner loop commonly known as the **[j]**, it swaps the value until it repeatedly goes through the whole list. The loop will continue to iterate as the sorting comparing and sorting continues until no more swaps are initiated.

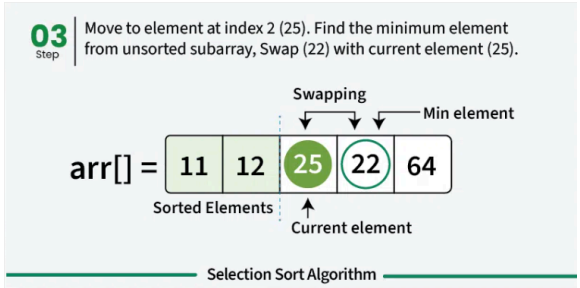
4. Give sample programs in C++ that use sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.

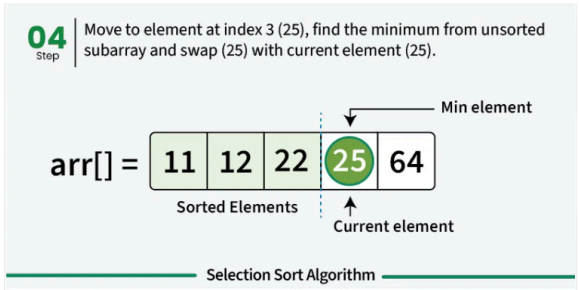
Sorting Algorithms	VISUAL REPRESENTATION	Explanation
Bubble Sort		This sorting algorithm works by iterating over and over again the adjacent elements if the elements are in the wrong order. This algorithm is known for being inefficient for large data sets. In every iteration, it processes only those that have already not been moved to the correct position. This uses two for loops to compare numbers in the list. The first loop holds onto a position [i], starting from the very front. The second loop looks at the very next position [j] and checks all the numbers ahead. If the number at [i] is bigger than the number at [j], the program immediately swaps them. By comparing and swapping like this, the smallest numbers get pushed to the front and keep the bigger number compared to the next number one by one until everything is in order.

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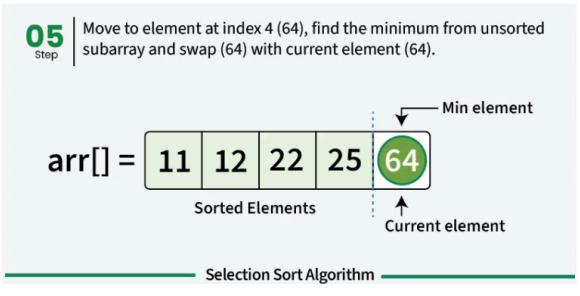
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<td>Selection Sort</td>
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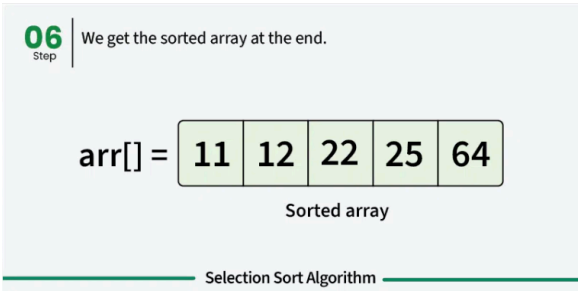


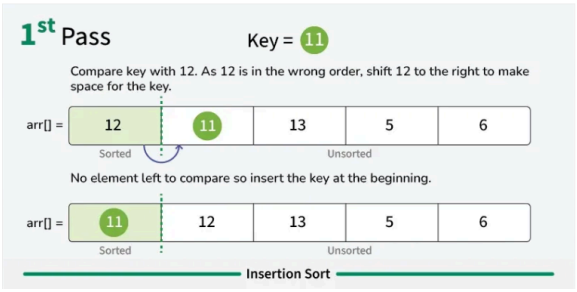


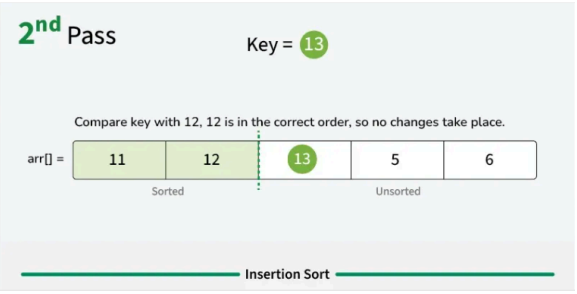


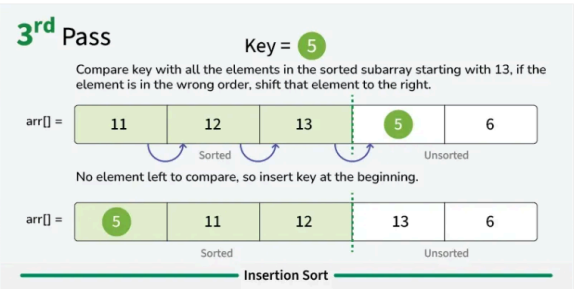
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<td>This Selection sort algorithm compares elements from the unsorted portion and swaps it with the first unsorted element. The goal of this algorithm is to sort the smallest element to its correct position by finding the smallest value among the elements and swap it with the second element. This uses a main loop and a helper function to sort the list. The main loop starts at the beginning of the list and goes through it step by step. For every step, it calls the helper function to look at the rest of the unsorted numbers and find the exact spot of the smallest number. Once it finds that smallest number, it uses std::swap to switch it with the number at the current position[i]. This puts the smallest number in its correct place, and it repeats this until the whole list is sorted.</td>
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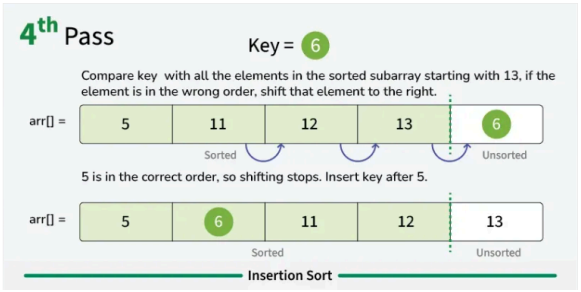


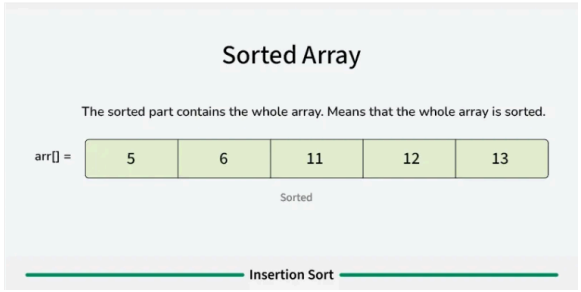
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<td>Insertion Sort</td>
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<td>The Insertion sort treats the current index as a sorted element as it hasn't fully checked the unsorted list. It is the same just like playing cards as stated during this. lesson. You split the cards into 2 parts, the sorted cards and the unsorted card. Then you pick a card from the unsorted group and you will insert that specific card right to its place in the sorted group. This sorts the list by moving numbers out of the way to make room for the current number. It uses a while loop that starts with the second number in the list and saves it in a temporary variable called temp. Then, an inner loop looks at the numbers before it. If temp is smaller than the number behind it, the program moves that larger number one spot to the right. It keeps doing this until it finds the right spot for temp, drops it in, and then moves on to the next number in the list.</td>
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REFERENCES:

GeeksforGeeks. (2026, June 2). *Bubble sort*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/bubble-sort-algorithm/

GeeksforGeeks. (2026a, February 24). *Insertion Sort Algorithm*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/insertion-sort-algorithm/

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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